Template Document

Academic Support Agent

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LaTeXAssignment .205 DSI205 Basics of Linear Algebra for Data Analytics

1 Latex PC

1. Install MiKTex `https://miktex.org/` TeXworks Editor
   MiKTex Compiler Latex PDF TeXworks Editor Latex
2. `https://www.overleaf.com/read/hyhynwsnpjdx`
   Menu > Download > Source
3. .tex TeXworks Compiler **XeLaTex** Menu
4. Compile/Typeset
5. Install Library
6. PDF

2 Overleaf Online

Latex *Overleaf* `https://www.overleaf.com`
: *Overleaf* Compile

1. Overleaf
2. `https://www.overleaf.com/latex/templates/dsi205-template/tsnyrzxnngdw` Open as Template (Overleaf)
3. Menu > Compiler Compiler **XeLaTex**
4. Recompile Auto-Compile (Auto Compile On)
5. Recompilie PDF Menu > Download > PDF

3

Math Mode 3

3.1 Inline mode

\(. . . \), $ . . . $, \begin{math} . . . \end{math}
$E = mc\hat{2}$ $E = mc^2$
\(E = mc\hat{2} \) $E = mc^2$

3.2 Displayed Mode Unnumbered

\[. . . \] $$ . . . $$
\[\bar{x} = \sum x_i \]

$$\bar{x} = \sum x_i$$

$$ \bar{x} = \sum x_i $$

$$\bar{x} = \sum x_i$$

3.3 Displayed Mode Numbered

\begin{equation} ... \end{equation}
\begin{equation}
I = \frac{V}{R}
\end{equation}

$$I = \frac{V}{R} \tag{1}$$

4

aligned equation

$$\begin{aligned} a &= 2 \times 2 \times b \times c \\ a &= 4 \times b \times c \\ a &= 4bc \end{aligned} \tag{2}$$

5

amsmath bmatrix equation

$$\mathbf{v} = \begin{bmatrix} 1 \\ 5 \\ -4 \\ 0 \end{bmatrix} \tag{3}$$

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 4 & 0 & 1 \end{pmatrix} \tag{4}$$

$$B = \begin{bmatrix} 1 & 2 & -3 \\ 4 & 0 & 1 \end{bmatrix} \tag{5}$$

$$c = \det A = \begin{vmatrix} -2 & 0 & 1 \\ 5 & -4 & 2 \\ 3 & 1 & 8 \end{vmatrix} \tag{6}$$